

UAV Software Ground School Notes (Week 2)

1. Introduction

- **How are drones used for aerial imagery?**
 - For search and rescue missions along with agriculture, law enforcement, and environmentalism
 - A ton of competitions!
 - Some aerial imagery!
 - Oblique (From a lower angle) vs Nadir (From straight up)
 - RGB-D/ multi-spectral/plain RGB
- Most recent drones utilized:
 - Gimbal Cameras (cheap, nice quality, easily usable)

2. How do we use OpenCV

- It is a generalist computer vision library that does a great many things.
- It is a Python/C++ free open-source library
- It covers:
 - Feature detection
 - 3D reconstruction
 - Motion analysis

- Optical flow
- Photo manipulation/de-blur/refining
- Video editing/reading/writing

A.What parts of OpenCV have we used?

- Feature detection (SIFT/ORB)
- Feature matching
- Image manipulations

B.SIFT vs Feature Matching

- SIFT can take two images and make 3D maps, it can stitch together these images to create a generalized map.
 - Ex: aerial photography over an area and creating a comprehensive map.

3. How can we do this? What are the drawbacks and Implications?

- OpenDroneMap workflows!

A.ODM benefits

- A lot easier to use

- Has really cool APIs/web application and contains 3d modelling
- You can even avoid coding entirely!
- It can give you incredible data!

B.ODM Drawbacks

- Not really optimized for being done in-flight
- Can be slow if not optimized beforehand
- Requires accurate GPS tags and runs poorly without it
- It is extremely feature rich with many add-ons for an edge device

4. What are we going to do now?

- This year we plan to either
 - Use a custom image stitching, SFM algorithm with OpenCV that can run real-time on the drone (done by Italy)
 - Really dig into OpenDroneMap specifics and find out how to divide it up into small subcomponents, also real-time in an instant
- A ton of cool potential projects too!
- SLAM (Simultaneous Localization/Mapping)
- Optical Flow
- Structure from Motion (getting a terrain map from image data)
- Applying drone imagery